VLM-DENTAL system architecture

an agentic tool loop over native-resolution radiographs, trained on verified self-correcting traces

Data layer

2 datasets in use: DENTEX · Tufts
(8 more catalogued as candidates - pending)

loaders: DENTEX full · Tufts partial (1 open question)
FDI notation conversion at ingest



panoramic radiograph + per-tooth boxes + diagnoses

multi-finding ground truth
1-7 findings per image
full set - never truncated

grounding model: YOLOv8m · 5-fold CV · mAP50 ~ 0.87

1

Trace synthesis (offline)

family A generates with-tools traces
+ paired no-tools traces

tiered perturbation of displayed boxes
(nudge behavior stays demonstrable)

family B verifies + repairs every trace
staged · cross-provider · multi-finding

dual-gate decontamination
regex sweep + re-verification

corpus: 1,847 + 1,847 · 3,694 traces
exact per-cohort parity

details in the trace-synthesis figure
(all quotes, boxes and stats are real)

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Training

1 · SFT on verified traces
behavior cloning of the tool loop
(1,847 with-tools traces)

then 3

2 · GRPO on fresh rollouts
same loop, live tool execution
group-relative advantage

accuracy · format · tool use · efficiency

efficiency term exempts locate_tooth + nudge_crop

output: LoRA adapters
loaded by the policy at inference

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rollouts execute the real dispatcher - same code path as trace generation
checkpoint selection on the dev split
(test set evaluated once, at the end)

Evaluation

set-matched precision / recall / F1
over multi-finding predictions

95% bootstrap confidence intervals

probes (mechanism evidence):
- tool removal: with vs without
- perturbation: shown-box vs true-box
- nudge budget K sweep

(results pending, reported as open cells)

heavy runs on GPU or TPU
(target decision pending - see input geometry)

5 per-image findings

dev-split signal 6

final answer - findings JSON

```
{"findings":[{"quadrant":3,"tooth":34,
"diagnosis":"Caries","confidence":0.82}]}
```

on answer / budget

Policy

Qwen3.5-9B

reasoning node
thought + JSON tool calls

LoRA adapters:
SFT · GRPO (pluggable)

call

obs

Tool node



live crop

8 registered tools

zoom · window · denoise
enhance · contrast
locate · nudge
bilateral compare · fdi

continuous parameters:
strength · center · width



input · native OPG (2,808 × 1,316 px)
every tool executes at full resolution

base_image

shared dispatcher - every image-consuming tool runs on base_image;
the crop chain never compounds

input geometry: TPU = canonical views + static slots (no recompile);
GPU = native so far - decision pending; tools always run native

numbered path: 1 data -> traces · 2 corpus -> SFT · 3 SFT -> GRPO · 4 adapters -> policy · 5 findings -> eval · 6 dev-split signal